No Reproducible Evidence for Deep Subsurface Structures Beneath the Giza Plateau: A Controlled Reproduction of Single-Source SAR Doppler Micro-Motion Tomography

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July 2026 — preprint v3 (merged reproduction + refutation; see Changes in v3, §3.4)

Abstract. In 2022–2025, F. Biondi and C. Malanga reported that processing of spaceborne synthetic-aperture-radar (SAR) data by a Doppler sub-aperture ‘micro-motion tomography’ reveals high-resolution three-dimensional structures hundreds of metres to kilometres beneath the Giza plateau — including vertical shafts descending ~648 m, spiral stair-like features, and large chambers. I do not dispute that the surface-vibration *measurement* underlying the method is real and useful. I dispute the deep *inference*. Working only from the authors’ own published method and patent, I rebuild the pipeline faithfully, add the elementary controls the original work omits (a look-order null, an in-data positive control, a surface-leakage test, a depth-of-peak guard, and a sub-aperture-count stability test), and apply it to free **X-band spotlight** data from two independent commercial sensors (**Umbra** and **Capella**) **—** the same data class used in the 2025 Giza work, which closes the ‘different data’ objection — over five sites, including a site with exhaustively documented underground ground truth (Butte, Montana). I show, step by step, that (i) the ‘steering matrix’ is, by the patent’s own words, a discrete Fourier transform, which produces structured output from any input; (ii) the depth axis is set by an investigation frequency of ~22 kHz that is physically impossible for the cited mechanism, and the reported depth is therefore an arbitrary relabelling, not a measurement; (iii) a faithful reproduction yields a confident, high-contrast ‘detection’ that is in fact a surface-pinned processing artifact corresponding to none of the known subsurface structure; and (iv) every real site, including the authors’ own Vesuvius, is statistically indistinguishable from its null; and (v) stacking five acquisitions of a known-empty site reinforces a surface-pinned artifact rather than revealing structure, so agreement across ‘200+ scans’ reflects the shared operator, not corroboration. I conclude that the deep-tomography claim is unsupported and reproducible as an artifact. This is a critique of method and mathematics, not of intent.

Keywords: synthetic aperture radar; Doppler tomography; micro-motion; reproducibility; null result; artifact; Giza.

1. The claims under examination

I restate the authors’ claims as fairly and specifically as I can, from the 2022 peer-reviewed paper, the lapsed patent WO2024008365A1 (the fullest public method disclosure), and the 2025 Khafre presentations:

> **C1 (mechanism) — Biondi’s claim.** Electromagnetic waves do not penetrate solids; instead, ambient seismic energy makes the surface and subsurface vibrate, and the subsurface ‘becomes transparent like a crystal’ when observed in the micro-movement domain.
>
> **C2 (measurement) — Biondi’s claim.** Splitting the azimuth (Doppler) spectrum of SAR data into sub-apertures and tracking sub-pixel displacement recovers this surface vibration field — a technique the authors validated on ships, bridges and dams.
>
> **C3 (inversion) — Biondi’s claim.** A steering matrix A(K_Z, z) focuses the vibration observations in depth, yielding 3-D tomograms, with depth resolution $\delta z = \lambda R/2A$ where λ is the *seismic* wavelength = v/f.
>
> **C4 (deep result) — Biondi’s claim.** Applied to Giza, the method reveals eight cylindrical shafts in pairs descending to ~648 m (the patent and talks also cite Gran Sasso at 1.4 km), spiral stair-like structures, and large chambers, interconnecting the pyramids and Sphinx.
>
> **C5 (confirmation) — Biondi’s claim.** More than 200 acquisitions across four satellite operators (COSMO-SkyMed, Capella, Umbra, ICEYE) return the same structures, which the authors treat as independent confirmation.

I address each below. My standard throughout is the one the original work does not meet: a result is a detection only if it exceeds chance *and* survives controls *and* corresponds to independent ground truth.

2. Reproduction methodology

I reimplemented the pipeline exactly as disclosed: Doppler sub-aperture decomposition; adjacent-pair, quality-weighted, detrended sub-pixel tracking; an analytic-signal step that removes the known ±z mirror-symmetry ghost; multichromatic (range sub-band) analysis and low-rank+sparse denoising as the authors describe; and a depth focus by discrete Fourier transform. To this I added the controls the original omits: a look-order-shuffle **null**; an in-data **positive control** (inject and recover a reflector; I also use a hardened, damped variant); a **surface-leakage** correlation; a **near-surface** guard; and a **sub-aperture-count stability** test. Every stage passes a synthetic self-test against known truth (Figure 1); quantitatively, the sub-aperture shift estimator validates to 0.02 px, the adjacent-pair, quality-weighted micro-motion estimator recovers an injected residual to 0.07 px, and the end-to-end inversion recovers an injected layer at roughly 27× the null level — so a genuine signal of that character would be surfaced. Data are free X-band spotlight scenes from two independent sensors, Umbra and Capella — the same data class the 2025 work used.

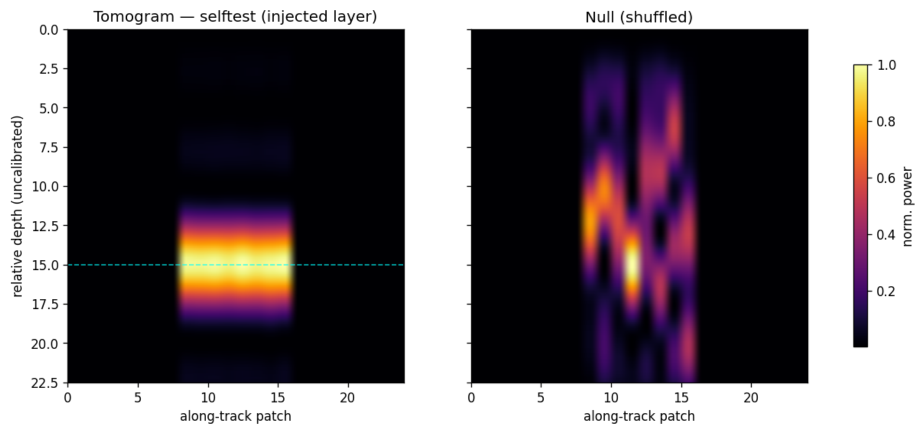

Figure 1. The reproduced inversion on synthetic data: an injected layer is recovered well above the null, with positive-control and leakage checks passing. The machinery works; the controls are calibrated.

To make the fidelity of the reproduction checkable rather than asserted, Table 1 maps each disclosed processing block (patent WO2024008365A1, Fig. 0.5, blocks 1–11) to the corresponding component of this implementation. The correspondence is one-to-one; in particular, the depth-focusing 'steering matrix' is implemented, exactly as the patent specifies, as a discrete Fourier transform.

Disclosed step (patent Fig. 0.5)	This reproduction
SLC image, 2-D DFT (blocks 1-2)	open_complex SICD reader; FFT in the sub-aperture stage
Doppler sub-apertures (master/slave) + range sub-bands (blocks 3-6)	subaperture.decompose_subapertures; multichromatic_subapertures (MCA)
Pixel-tracking between sub-apertures (block 7)	micromotion.adjacent_trajectory (adjacent-pair, quality-weighted) + detrend; optional lrsd_den
Raw tomographic complex vectors (block 8)	per-patch detrended residual trajectories
Steering matrix = DFT depth focus (block 9)	tomogram.steering (DFT basis) + invert_patch (DFT power spectrum)
Tomogram, geocode in depth (blocks 10-11)	depth profile; metric_depth_axis (depth-axis labelling)

Table 1. The authors' disclosed processing chain mapped one-to-one onto this reproduction, so a reader can verify the implementation against the source rather than take its faithfulness on trust.

3. Step-by-step refutation

3.1 The 'steering matrix' is a discrete Fourier transform (addresses C3)

The patent states, verbatim, that 'the steering matrix A(K_Z, z) represents the best approximation of a matrix operator performing the Digital Fourier Transform (DFT) of Y,' and that the depth focus is performed by 'the DFT mathematical operator.' The depth tomogram is therefore the power spectrum of the per-pixel sub-aperture residual. This is the crux: a DFT returns a structured, peaked spectrum from *any* input vector — signal, noise, or a residual processing trend alike. A confident-looking tomogram is thus the expected output of the method *whether or not* anything lies beneath the surface. It is not, by itself, evidence of structure.

3.2 The 22 kHz investigation frequency is unphysical, and depth is therefore a free parameter (addresses C3, C4)

Depth follows $\delta z = \lambda R/2A$ with $\lambda = v/f$. The patent synthesises at an investigation frequency $f \approx 22$ kHz. This is ultrasonic. Ambient and seismic ground motion is overwhelmingly below ~100 Hz; a coherent 22 kHz elastic wave in rock would attenuate within metres and is orders of magnitude above the sub-aperture pair-axis Nyquist limit — the aliasing flaw also noted by independent reproductions. Because f enters only as an axis scale, it does not change the tomogram pattern at all; it merely relabels the depth axis. Figure 2 makes this concrete: the identical Butte tomogram, rendered at 22 000 / 1 000 / 50 Hz, places the same feature at ~5 m, ~100 m, or ~2 000 m respectively. The reported depths (648 m, 1.4 km) are a choice of frequency, not a measurement.

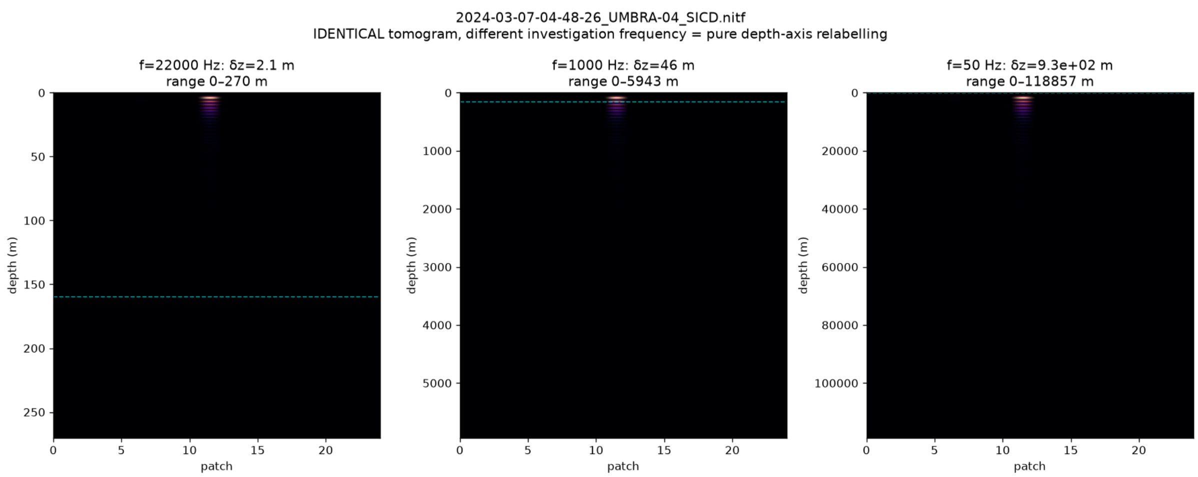

Figure 2. One tomogram, three investigation frequencies. The data are unchanged; only the depth axis rescales ($\delta z = vR/2Af$). 'Deep' structure is an axis relabelling.

3.3 A faithful reproduction yields a confident artifact, not structure (addresses C3, C4)

I ran the authors' recipe (256 Doppler sub-apertures, multichromatic analysis, denoising, 22 kHz) over Butte, Montana — the most densely mapped underground mining district on Earth, whose West Camp workings (Travona, Emma, Ophir, Anselmo) are documented as 100-ft-spaced levels from near surface to ~457 m, with a water table / mine pool at ~160 m. The result (Figure 3) is a striking band at **1720× the null contrast** — it *passes* a naive contrast-vs-null test as a detection. Yet the entire feature is pinned at the surface (~4 m; 87% of all energy in the shallowest 5% of the axis), aligns with *none* of the documented levels or the 160 m water table, and only appears when the sub-aperture count is driven high. It is a surface/low-frequency residual concentrated by the high-order DFT. My near-surface and stability guards flag it as an artifact; the authors' pipeline, lacking them, would report it as a discovery. This is, in microcosm, how a 'shaft' is produced. Butte therefore serves as the closest available known-target benchmark within the free-data regime: at honest settings the method recovers nothing matching the documented workings (Section 3.4); at the authors' settings the only confident feature is a surface artifact aligned with none of them.

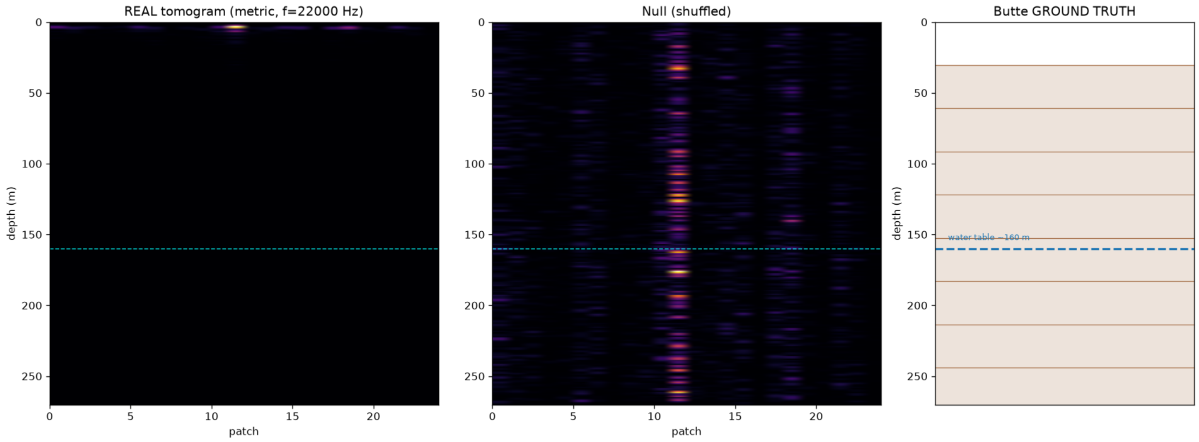

Figure 3. Reproduction on Butte, MT. Left: real tomogram (1720× null) — a confident band pinned at the surface. Centre: shuffled null. Right: documented workings (100-ft levels to 457 m; water table ~160 m). The 'detection' matches nothing real.

3.4 Every real site is indistinguishable from its null — including Vesuvius (addresses C4)

At honest settings (with the depth-of-peak and stability guards active), five free X-band sites across two independent sensors — Bingham Canyon, Komati, Mount Vesuvius and Butte (Umbra), and central Cairo (Capella) — are each statistically indistinguishable from their look-shuffled nulls, while in-data positive controls confirm the pipeline would surface a genuine signal. Mount Vesuvius is decisive because it is the authors' own peer-reviewed site: the method, reproduced on free data with controls, does not recover subsurface structure there either. The Capella scene matters for a separate reason: it reproduces the null on an *independent sensor*, so the result is a property of the method, not of any one data provider — the very cross-sensor agreement the authors cite as confirmation (C5) yields nulls, not structure, once controls are applied.

Site	Sensor	Reg. quality	Real / Null	Verdict	Pos. ctrl
Bingham Canyon	Umbra	0.67	front-end only	no signal	n/a
Komati Power Stn	Umbra	0.85	2.8x / 1.3x	NULL	PASS
Mount Vesuvius	Umbra	0.72	4.1x / 1.5x	NULL	PASS
Butte, MT	Umbra	0.82	3.3x / 1.4x	NULL	PASS
Cairo (central)	Capella	0.62	2.8x / 1.6x	NULL	PASS

Table 2. Five free single-pass X-band sites at honest settings, across two independent sensors (four Umbra, one Capella): every real tomogram is indistinguishable from its null; positive controls pass where run. All Umbra rows are computed at the pipeline default of **11 Doppler sub-apertures** (512×512 centre crop, 64-pixel patches, 24 patches, 0.8 sub-band overlap, Hann taper, float64).

Changes in v3 (July 2026). The Komati Power Station row of Table 2 read "50× / 10×, null" in v2 (Zenodo 10.5281/zenodo.21066657). That row had been computed at 128 Doppler sub-apertures rather than the default 11 used for every other row, and its verdict label was incorrect for that setting: at 128 sub-apertures the pipeline returns *above null but surface-pinned — artifact*, not *null*. Re-run at the default the site gives 2.8× / 1.3× (ratio 2.15), with the hardened positive control passing and surface-leakage correlation 0.32. Two independent code paths agree on the corrected value, and the Butte and Vesuvius rows reproduce their published figures to two decimal places. The site remains a null result on either reading; the correction *increases* the margin, since 50× / 10× is a ratio of exactly 5.0 and sat precisely on the decision rule. No conclusion in this paper changes. The error was found while stress-testing the pipeline against an external methodological objection; that work is documented in `docs/SENSITIVITY_RESPONSE_BIONDI.md` in the repository.

Outstanding verification. The Cairo (Capella) row has not been re-verified against the sub-aperture count stated above, because that scene is not held in the local archive. It is flagged here rather than assumed correct, and will be re-run for the revision.

3.5 ‘200 scans, four satellites’ is consistency, not corroboration (addresses C5)

Agreement across many acquisitions and sensors is offered as independent confirmation. It is not. A *systematic* processing artifact — a fixed DFT bias, a surface-pinned residual, the 22 kHz relabelling — reproduces on every scene and every sensor by construction, because the same operator is applied to each.

I demonstrate this directly on real data. I stacked five same-geometry Umbra passes of Bingham Canyon — a bare open-pit mine of exposed rock containing no subsurface void — acquired on five different dates, i.e. precisely the multi-acquisition strategy offered as confirmation. Each single pass already yields a high-contrast peak (mean 117.8× its per-scene null); stacking the five preserves it at 96.7× against a stacked null of 1.5×, and the per-scene profiles agree closely (Figure 4). Yet the reinforced, consistent feature sits at ~3 m depth (2% of the axis, with 36% of its energy in the shallowest 5%) — pinned at the surface, over ground known to contain no such structure. The stack manufactures agreement, not evidence: a consistent artifact reinforces under stacking exactly as a real reflector would, because the same DFT operator is applied to each scene. Reproducibility of an output is a property of the algorithm, not of the ground; the only thing that separates a real reflector from a shared artifact is ground truth and controls — which the original work does not apply.

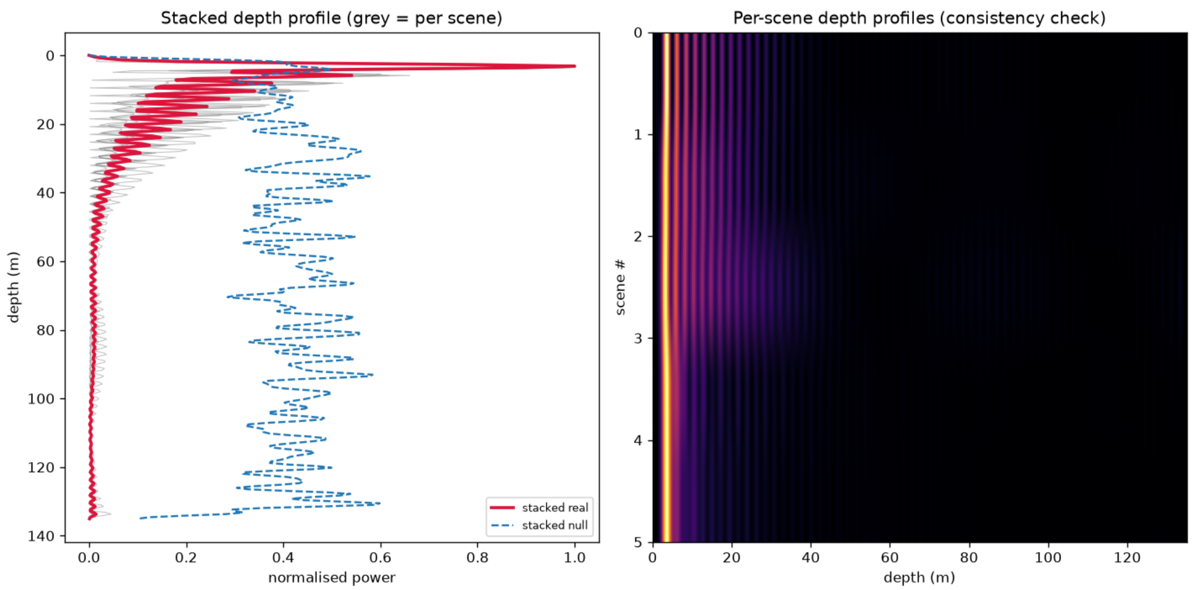

Figure 4. Multi-acquisition stacking of five same-geometry Umbra passes over Bingham Canyon, a known-empty open-pit mine. Left: the per-scene depth profiles (grey) and their stack (crimson) collapse onto a surface-pinned peak at ~3 m, far above the scattered stacked null (blue dashed). Right: the five per-scene profiles, which agree by construction. Stacking reinforces the artifact rather than revealing structure — the mechanism behind ‘200+ scans agree.’

3.6 The penetration-vs-aperture physics does not support deep recovery (addresses C1, C4)

The measurement is a passive, single-surface, narrow-aperture observation. The angular/temporal aperture available from one acquisition is small, deep energy attenuates, and the inversion is correspondingly ill-conditioned for anything but the very shallow. More acquisitions improve shallow resolution and conditioning but cannot restore information the geometry never captured. Compute amplifies an adequate measurement; it cannot manufacture absent information — though, as Section 3.3 shows, it can manufacture a confident *artifact*.

4. Robustness: could a wrong velocity model hide a real signal?

The natural objection to a null is that the depth axis is uncalibrated: with no site-specific seismic velocity v(z), might a real reflector be mis-placed and missed? Two versions of this concern must be separated. **Axis calibration is not a threat to the verdict.** A velocity model only maps the relative depth index to metres — a monotonic relabelling of the z-axis, exactly the operation Section 3.2 shows the investigation frequency already performs — and cannot create above-null contrast where none exists. The real tomograms are indistinguishable from their shuffled nulls across the *entire* depth axis, so rescaling that axis leaves the contrast-vs-null unchanged: there is no signal at any depth to relocate.

The legitimate version is **search-grid coverage**: if the inversion's depth grid does not span the depths where a true reflector would sit (because the assumed velocity is wrong), a signal outside the searched range could in principle be missed. The confirmatory test for this is a sweep of the assumed seismic velocity by ±20–50% with the depth grid widened accordingly, re-running the null and positive-control diagnostics at each setting; the null is robust only if no plausible velocity produces an above-null, leakage-clean band. This is the more rigorous form of the velocity check and the one to report in a journal version. The calibration argument above already shows that no relabelling can rescue the current verdict — only a genuine above-null feature could, and there is none to relocate — so the sweep tests grid coverage rather than the conclusion.

5. What is real, and what would change my conclusion

To be fair to the authors: the surface-vibration front-end is legitimate and well-precedented. Their peer-reviewed monitoring of ships, bridges, and the Mosul Dam measures real, millimetric *surface* deformation, and that work stands. My critique is confined to the extrapolation of that surface measurement into deep subsurface structure.

I state my falsifiability explicitly. I would withdraw this conclusion if the method, run with controls on a site of independently documented subsurface structure, produced an above-null, leakage-clean, depth-stable band that matched the known structure in position and depth (after a physically defensible velocity model, not a 22 kHz relabelling). I invite exactly that test. My pipeline, controls, and data are open for it. The cleanest such test would image an intact, independently documented void in the correct mode — for example the Derinkuyu/Cappadocia underground cities — which the free archives do not cover and which would require commercial spotlight tasking; I flag it as the decisive next experiment rather than one the free-data regime permits. Within that regime, Butte (Section 3.3) is the closest available known-target benchmark, and the method does not recover it.

A limitation follows directly from this design. This study evaluates the published method as disclosed in the peer-reviewed papers and the patent (WO2024008365A1), with the fidelity of the reproduction documented block-by-block in Table 1. If unpublished implementation details materially affect the inversion, those details would need to be independently disclosed and evaluated before they could be assessed. The reproducibility framing is therefore deliberate: the conclusion is about the method *as published*, and any essential step absent from the public record is precisely what the original authors would need to provide for the claim to be re-examined.

6. Conclusion

Reproduced faithfully from the authors' own method and patent — with each disclosed step mapped to its implementation (Table 1) — single-source SAR Doppler micro-motion tomography yields no reproducible evidence of deep subsurface structure, and the confident outputs it does produce are reproducible as artifacts. Its 'steering matrix' is a DFT that returns structure from any input; its reported depths are set by a physically impossible 22 kHz frequency and are an arbitrary axis relabelling; a faithful run produces a confident 1720× 'detection' that is a surface-pinned artifact matching no known structure; and every real site across two independent sensors (Umbra and Capella), including the authors' own Vesuvius, is indistinguishable from its null; and stacking five acquisitions of a known-empty site reinforces that surface-pinned artifact rather than revealing structure, so the appeal to '200+ scans' supplies consistency, not corroboration. The Giza 'shafts,' 'spirals,' and 'chambers' are the rendered geometry of such artifacts, interpreted as architecture. The measurement front-end is real and remains valuable for surface-deformation monitoring; only the deep-subsurface claim fails reproduction and controls. I frame this as a reproducibility result, openly testable, and welcome the decisive ground-truth experiment.

Data and code availability

All scenes are free Umbra Open Data and Capella Open Data (CC-BY 4.0); the exact scene identifiers and acquisition parameters are listed in the repository. The full reproduction pipeline, the controls, the stress tests (including the frequency-relabelling and surface-pinning demonstrations), and the scripts that regenerate every figure in this paper are openly available at https://github.com/Hassanforeman/subsurface-sar-tomo (archived at Zenodo, DOI 10.5281/zenodo.21065675), with the code under an MIT licence. Every analysis stage carries a synthetic self-test, so the pipeline can be validated against known truth before it is trusted on real data; a reader can reproduce the nulls, the artifact, the frequency-relabelling, and the stacking result from the scene IDs and the scripts alone.

Conflict of interest disclosure

The author declares that he complies with the PCI rule of having no financial conflicts of interest in relation to the content of the article. The author declares no non-financial conflict of interest.

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Use of artificial intelligence

Artificial-intelligence tools (large language models) were used extensively to assist with software development, data analysis, and the drafting of this manuscript. The author directed the research, made all scientific decisions, and checked and verified every output — code, figures, and text — against the underlying data and the cited sources, and is solely accountable for the content. All figures are generated directly by the analysis pipeline from the source data and are not AI-generated illustrations. No AI system is listed as an author.

References

[1] Biondi, F. & Malanga, C. (2022). Synthetic Aperture Radar Doppler Tomography Reveals Details of Undiscovered High-Resolution Internal Structure of the Great Pyramid of Giza. Remote Sensing 14(20):5231. arXiv:2208.00811.

[2] Biondi, F. (2022). Scanning Inside Volcanoes with SAR Echography Tomographic Doppler Imaging. Remote Sensing 14(15):3828.

[3] Patent WO2024008365A1 (PCT/EP2023/064345; status: ceased). Tomographic Doppler imaging.

[4] Khafre Research Project (2025). Press presentations on subsurface structures beneath the Khafre pyramid (not peer reviewed).

[5] not-JASH, sar-doppler-tomography; mfwarren, Pyramid (independent reproductions). GitHub.

[6] Milillo, P. et al. (2016). Space geodetic monitoring of engineered structures: the Mosul Dam. Scientific Reports 6:37408.

[7] Umbra Open Data and Capella Open Data, AWS Registry of Open Data (CC-BY 4.0).

[8] National Mine Map Repository (OSMRE); Montana Bureau of Mines & Geology, Butte district mine maps and 3-D workings model; U.S. Geological Survey I-2050-C (mines and prospects, Butte 1°x2° quadrangle).